Goose neck tool

This is a special type of tool used for finish machining. Very good surface finish will be obtained. The cutting edge of goose neck tool has a springy action. The tip of the cutting edge lies in the same line with the rear side of the shank.

Slot cutting tool

Wide rectangular or square grooves are known as slots. Rough machining of the slot is carried out using round nose tool. After that, a slot cutting tool is used for finish machining work.

T-slot cutting tool

The central rectangular (or square) slot is first machined using rough machining tool and then by using parting tool (or slot cutting tool). After that, a T-slot cutting tool is used to machine underneath the rectangular groove.

Form tool

Form tools are made to suit some specific requirements for machining V shaped grooves or similar special shaped grooves in concave or convex form.

3.13 Setting of shaper table, vise and toolhead

The machining accuracy will not be perfect if the machine table and toolhead are not set properly. When the sides are perpendicular, it is referred as squareness. Parallelism means the two sides are absolutely parallel to each other. Alignment is an arrangement in which the relative positions of the table, the jaws of the vise, the toolhead and the ram are perfect. The above setting of the table, work and the tool are done with the help of test bars and feeler gauges.

3.14 Types of shaping operations

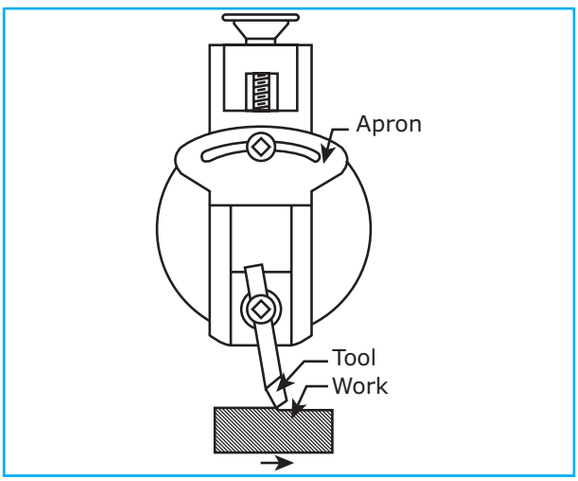
Different types of operations are performed in a shaping machine. They are broadly classified as

1. Regular operations
2. Special operations

General operations

Machining horizontal surfaces

A shaper is mostly used to machine a flat, true surface on a workpiece. Horizontal surfaces are machined by moving the work mounted on the machine table at a cross direction with respect to the ram movement. The clapper box can be set vertical or slightly inclined towards the uncut surface. This arrangement enables the tool to lift automatically during the return stroke. The tool will not drag on the machined surface.



Machining Horizontal Surfaces

Machining vertical surfaces

A vertical cut is made while machining the end of a workpiece, squaring up a block or machining a shoulder.

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